

ESE 101 Homework 2

Tia Abraham

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1 Jupiter

Jupiter's mean distance from the sun is 7.8×10^8 km (or 5.2 Astronomical Units). Its (Bond) albedo is 0.34, and the solar luminosity is 3.8×10^{26} W.

Part 1

What is the total solar irradiance at Jupiter?

The total solar irradiance S at a distance d from the sun is given by the inverse square law (*Physics of Earth's Climate* Eq. 3.12, Radiative Energy Slides page 3), which distributes the total solar luminosity $L_{\odot}$ over the surface of a sphere:

$$S = \frac{L_{\odot}}{4\pi d^2}$$

We are given the following values for Jupiter:

- $L_{\odot} = 3.8 \times 10^{26}$ W
- $d = 7.8 \times 10^8$ km = 7.8×10^{11} m

Substituting these values into the equation:

$$S = \frac{3.8 \times 10^{26} \text{ W}}{4\pi(7.8 \times 10^{11} \text{ m})^2}$$

$$S \approx 49.7 \text{ W m}^{-2}$$

Part 2

In your own words (no math), explain brightness temperature and effective temperature to your college-level friend who is not taking this class. Make sure to mention the relationship between the two of them. Feel free to add drawings if useful to your explanation.

Effective Temperature

Effective temperature is the theoretical temperature a planet would need, if it were a perfect blackbody, to radiate the same total amount of energy that it absorbs from the sun. It is a single, calculated value that represents the planet's overall energy balance.

Brightness Temperature

Brightness temperature is the temperature a perfect blackbody would need to have to match the intensity of radiation measured from the planet at one specific wavelength of light (**Radiative Energy Slides page 14**). A planet has many different brightness temperatures, one for each wavelength you measure, which reveals the temperature of the specific layer (surface or high atmosphere) that is emitting that light.

- In an atmospheric “window” (a transparent wavelength), the brightness temperature is high, as we are seeing the warm surface (**Radiative Energy Slides page 9, “288 K” curve**).
- In a strong absorption band (like for CO_2), the brightness temperature is low, as we are seeing the cold upper atmosphere (**Radiative Energy Slides page 9, “ CO_2 ” dip**).

The Relationship

The relationship is about the total versus the parts. The effective temperature represents the total energy the planet radiates out to space, combining all the different wavelengths into a single, equivalent temperature. The brightness temperature describes the energy radiated at one specific wavelength at a time. If you were to measure all the individual brightness temperatures across the entire spectrum and add up the energy they all represent, that total combined energy is what the effective temperature describes.

Part 3

What would be the effective temperature of Jupiter if it were a perfect black-body (in the longwave band) in equilibrium with the insolation?

To find the effective temperature (T_e), we assume the planet is in radiative equilibrium, where the absorbed incoming solar (shortwave) energy equals the emitted outgoing (longwave) energy.

The absorbed energy is the incoming solar irradiance (S), minus the reflected portion (given by the albedo α), multiplied by the cross-sectional area of the planet (πr^2):

$$E_{in} = S(1 - \alpha) \cdot \pi r^2$$

The emitted energy is given by the Stefan-Boltzmann law for a blackbody, multiplied by the entire surface area of the planet ($4\pi r^2$):

$$E_{out} = \sigma T_e^4 \cdot 4\pi r^2$$

Setting $E_{in} = E_{out}$:

$$S(1 - \alpha)\pi r^2 = \sigma T_e^4 4\pi r^2$$

The πr^2 terms cancel, giving the standard equation for effective temperature (**Physics of Earth's Climate Eq. 3.47**):

$$\frac{S(1 - \alpha)}{4} = \sigma T_e^4$$

Solving for T_e :

$$T_e = \left(\frac{S(1 - \alpha)}{4\sigma} \right)^{1/4}$$

We use the following values:

- $S \approx 49.7 \text{ W m}^{-2}$ (from **part 1**)
- $\alpha = 0.34$ (given)
- $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ (Stefan-Boltzmann constant, **Physics of Earth's Climate Ch. 3**)

Substituting these values:

$$T_e = \left(\frac{49.7 \text{ W m}^{-2} \cdot (1 - 0.34)}{4 \cdot 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}} \right)^{1/4}$$

$$T_e \approx 109.7 \text{ K}$$

Part 4

The measured effective temperature of Jupiter is 125 K. How large is the additional energy flux that must be emanating from Jupiter's interior to lead to this effective temperature? *[This energy flux is somewhat analogous to the (small) geothermal energy flux on Earth, but it is much larger and arises through different processes: It arises primarily because Jupiter is still contracting, and the gravitational energy that is released by the contraction emanates as an internal energy flux.]*

The measured effective temperature (T_{measured}) of 125 K accounts for all energy radiated by the planet. This total outgoing flux (F_{out}) must be balanced by the sum of the absorbed solar energy (F_{in}) and the internal heat flux (F_{internal}).

$$F_{\text{out}} = F_{\text{in}} + F_{\text{internal}}$$

The total outgoing flux is given by the Stefan-Boltzmann law using the measured temperature:

$$F_{\text{out}} = \sigma T_{\text{measured}}^4$$

The absorbed solar energy is the term we calculated in **part 3**, which gave us the equilibrium temperature T_e :

$$F_{\text{in}} = \frac{S(1 - \alpha)}{4} = \sigma T_e^4$$

By rearranging, the internal flux is the difference between the total flux radiated and the flux from absorbed sunlight:

$$F_{\text{internal}} = F_{\text{out}} - F_{\text{in}} = \sigma T_{\text{measured}}^4 - \sigma T_e^4 = \sigma(T_{\text{measured}}^4 - T_e^4)$$

Using the values:

- $T_{\text{measured}} = 125 \text{ K}$ (given)
- $T_e \approx 109.7 \text{ K}$ (from **part 3**)
- $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Substituting these values:

$$F_{\text{internal}} = (5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4})((125 \text{ K})^4 - (109.7 \text{ K})^4)$$

$$F_{\text{internal}} \approx 5.64 \text{ W m}^{-2}$$

2 Insolation

Write a program that calculates the diurnally averaged insolation as a function of time t given the orbital parameters obliquity γ , longitude of perihelion ϖ , and eccentricity e . (See the book draft for how to calculate insolation.) If you're interested in higher order expansions or variations in orbital parameters consider referencing the source code used in the CliMA earth system model (<https://github.com/CliMA/Insolation.jl/>). With your program, please do the following:

```
1 # Imports
2 import numpy as np
3 import matplotlib.pyplot as plt
```

Part 1

Calculate and plot the insolation as function of latitude at the top of Earth's atmosphere for each day of 2025.

```
1 def calculate_insolation_for_year(
2     year=2025, e=0.017, omega_deg=282.95, gamma_deg=23.44):
3     """
4     Calculates the diurnally averaged insolation for a given year and orbital
5     parameters.
6
7     This function follows the step-by-step procedure outlined in the
8     Insolation Calculation box in the Physics of Earth's Climate chapter 3,
9     page 95. It uses orbital parameters to calculate the daily variation of
10    insolation over all latitudes for a full year.
11
12    Args:
13        year (int): The calendar year for the calculation. This is used to
14                    determine leap years. Defaults to 2025.
15        e (float): Eccentricity of Earth's orbit. Defaults to 0.017, the value
16                  for 1/1/2000 from Table 3.1.
17        omega_deg (float): Longitude of perihelion in degrees. Defaults to
18                          282.95, the value for 1/1/2000 from Table 3.1.
19        gamma_deg (float): Obliquity of Earth's spin axis in degrees. Defaults
20                           to 23.44, the value for 1/1/2000 from Table 3.1.
21
22    Returns:
23        tuple: A tuple containing four elements:
24            - day_of_year (np.ndarray): 1D array of day numbers.
25            - latitudes_deg (np.ndarray): 1D array of latitudes from -90 to 90
26                                          degrees.
27            - insolation_grid (np.ndarray): 2D array (latitudes x days) holding
28                                              the calculated diurnally averaged
29                                              insolation in W/m^2.
30            - days_in_year (int): The total number of days in the given year.
31    """
32
33    # ----- Define Constants and Parameters -----
34    S0 = 1362.0 # Total Solar Irradiance (W/m^2)
35    ds = 1.000 # Semi-major axis (in AU)
36    Ya = 365.26 # Anomalistic year (days)
37
38    # Convert angles from degrees to radians for numpy functions
39    omega = np.radians(omega_deg) # Longitude of perihelion
40    gamma = np.radians(gamma_deg) # Obliquity
41
42    # Check for leap year
43    is_leap = (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0)
44    days_in_year = 366 if is_leap else 365
45
46    # March 21 is day 80 in a non-leap year, 81 in a leap year
47    day_of_vernal_equinox = 81 if is_leap else 80
```

```

48 # Create arrays for calculation
49 # 't_days' is days since vernal equinox, our reference time
50 day_of_year = np.arange(1, days_in_year + 1)
51 t_days = day_of_year - day_of_vernal_equinox
52
53
54 latitudes_deg = np.linspace(-90, 90, 181)
55 latitudes_rad = np.radians(latitudes_deg)
56
57 # 2D array to store results: rows=latitude, cols=day_of_year
58 insolation_grid = np.zeros((len(latitudes_rad), len(t_days)))
59
60 # ----- Calculate Parameters (Steps from p. 95) -----
61 # Calculate Mean Anomaly at Vernal Equinox (Mv0) -- Equation 3.11
62 beta = (1 - e**2)**0.5
63 Mv0 = -omega + (e + e**3 / 4) * (1 + beta) * np.sin(omega) \
64       - (1/2) * e**2 * (1/2 + beta) * np.sin(2 * omega) \
65       + (1/4) * e**3 * (1/3 + beta) * np.sin(3 * omega)
66
67 # Step 1: Calculate Mean Anomaly M(t) for all days -- Equation 3.10
68 M = (2 * np.pi * t_days / Ya) + Mv0
69
70 # Step 2: Calculate True Anomaly A(t) (to 3rd order) -- Equation 3.8
71 A = M + (2 * e - e**3 / 4) * np.sin(M) \
72       + (5 / 4) * e**2 * np.sin(2 * M) \
73       + (13 / 12) * e**3 * np.sin(3 * M)
74
75 # Step 3: Calculate Earth-Sun distance d(t) -- Equation 3.1
76 d_au = ds * (1 - e**2) / (1 + e * np.cos(A))
77
78 # Step 4: Get Solar Longitude Ls(t) -- Equation 3.9
79 Ls = A + omega
80
81 # Step 5: Calculate Declination Angle delta(t) -- Equation 3.16
82 delta = np.sin(gamma) * np.sin(Ls)
83
84 # ----- Loop through latitudes to complete calculations -----
85 for i, phi in enumerate(latitudes_rad):
86     # Step 6: Calculate Sunrise/Sunset Hour Angle (eta_d) -- Equation 3.19
87     tan_phi = np.tan(phi)
88     tan_delta = np.tan(delta)
89     cos_eta_d = -tan_phi * tan_delta
90
91     # Clamp values for polar day/night
92     eta_d = np.arccos(np.clip(cos_eta_d, -1.0, 1.0))
93
94     # Step 7: Calculate Diurnally Averaged Cosine of Zenith Angle
95     # -- Equation 3.20
96     cos_theta_bar = (1 / np.pi) * (eta_d * np.sin(phi) * np.sin(
97         delta) + np.cos(phi) * np.cos(delta) * np.sin(eta_d))
98
99     # Step 8: Calculate Diurnally Averaged Insolation F -- Pg. 95
100     S = S0 * (ds / d_au)**2
101     F = S * cos_theta_bar
102
103     insolation_grid[i, :] = F
104
105 return day_of_year, latitudes_deg, insolation_grid, days_in_year
106
107 # ----- Run Calculation and Plot -----
108 # We'll use the orbital parameters from 1/1/2025 from Table 3.1
109 # Orbital parameters vary very slowly over thousands of years (Milankovitch
110 # cycles), but the change from 2000 to 2025 is negligible for our purposes.
111 day_of_year, lats, insolation, days_in_year = calculate_insolation_for_year()
112
113 def setup_month_ticks(ax, days_in_year):
114     """Helper to add month ticks to a daily insolation plot."""
115     month_names = ['J', 'F', 'M', 'A', 'M', 'J', 'J', 'A', 'S', 'O', 'N', 'D']
116     if days_in_year == 366:
117         month_days = np.array([31, 29, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31])
118     else:
119         month_days = np.array([31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31])
120     month_centers = np.cumsum(month_days) - month_days / 2.0
121     ax.set_xlabel('Month')
122     ax.set_xticks(month_centers, month_names)
123     ax.set_yticks([-90, -60, -30, 0, 30, 60, 90],

```

```

124         ['90\u00B0S', '60\u00B0S', '30\u00B0S', '0\u00B0',
125          '30\u00B0N', '60\u00B0N', '90\u00B0N'])
126     ax.set_ylabel('Latitude')
127     ax.grid(axis='x', linestyle='--', alpha=0.5)
128
129 plt.figure(figsize=(10, 6))
130 # Found insolation levels for contouring using np.max and np.min
131 levels = np.arange(0, 551, 25)
132 # Plotting with contourf, similar to Fig 3.7
133 cf = plt.contourf(
134     day_of_year, lats, insolation, levels=levels, cmap='jet', extend='max')
135 cbar = plt.colorbar(cf)
136 setup_month_ticks(plt.gca(), days_in_year)
137 cbar.set_label('Insolation ($W\ m^{-2}$)')
138 plt.title('Daily Averaged Insolation (W/m\u00B2) for 2025')
139 plt.grid(axis='x', linestyle='--', alpha=0.5)

```

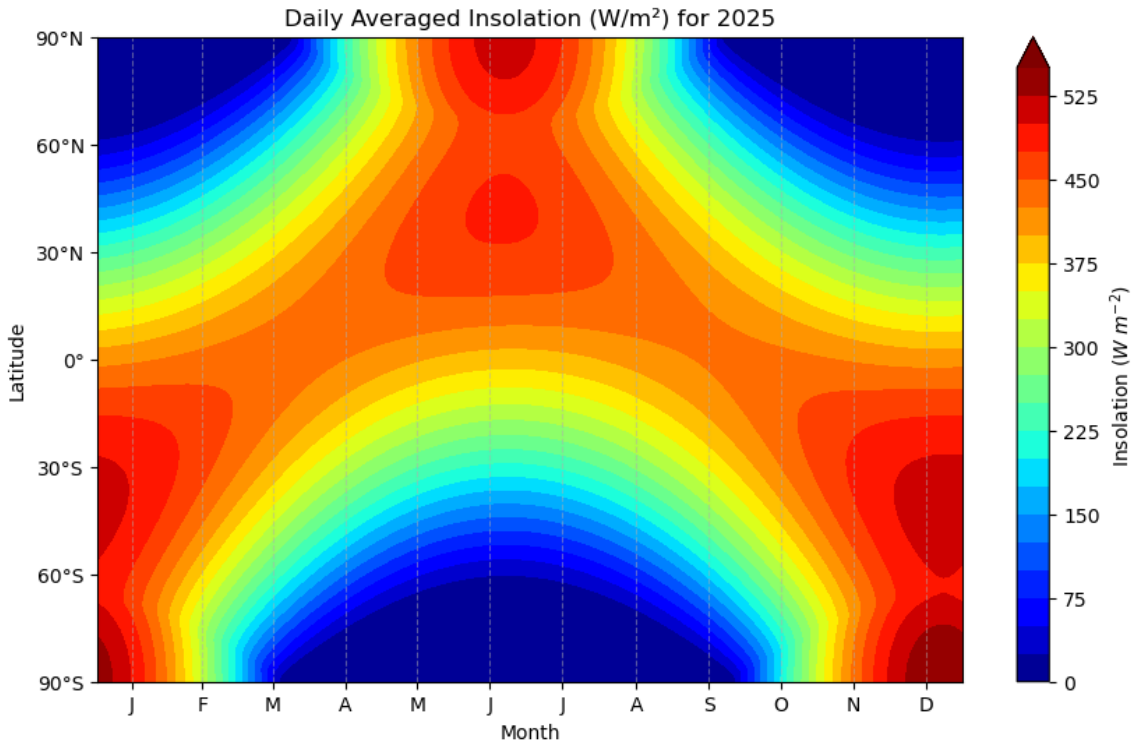


Figure 1: Diurnally averaged insolation ($W\ m^{-2}$) as a function of latitude and day of year for 2025 using present-day orbital parameters from J2000 (**Table 3.1**). Vernal equinox here is kept fixed at March 21.

Part 2

Calculate and plot the annually averaged insolation as a function of latitude.

```

1 # Average over the time axis (axis=1) of the insolation grid
2 annual_avg_insolation = np.mean(insolation, axis=1)
3
4 # Plot
5 plt.figure(figsize=(3, 7)) # Dimensions similar to Fig 3.7
6 plt.plot(annual_avg_insolation, lats)
7 plt.xlabel('Annually Averaged Insolation ($W\ m^{-2}$)')
8 plt.ylabel('Latitude')
9 plt.title('Annually Averaged Insolation (2025)')
10 plt.yticks([-90, -60, -30, 0, 30, 60, 90],
11            ['90\u00B0S', '60\u00B0S', '30\u00B0S', '0\u00B0', '30\u00B0N',
12             '60\u00B0N', '90\u00B0N'])
13 plt.ylim(-90, 90)
14 plt.grid(True, linestyle='--', alpha=0.7)

```

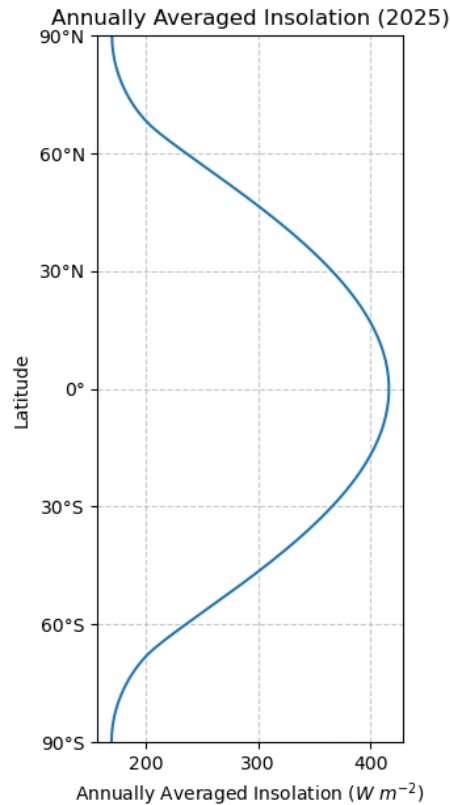


Figure 2: Annually averaged insolation (W m^{-2}) as a function of latitude, showing a maximum at the equator and minima at the poles.

Part 3.1

Turn the longitude of perihelion 180° compared to its present value. What is the date of perihelion now?

The present longitude of perihelion (ϖ) is 282.95° (*Physics of Earth's Climate*, Table 3.1). This angle is measured from the vernal equinox (March 21) to the point of perihelion (section 3.1). The current date of perihelion corresponding to this angle is January 3 (Figure 3.2). We are asked to turn the longitude of perihelion by 180° :

$$\varpi_{\text{new}} = \varpi_{\text{present}} + 180^\circ = 282.95^\circ + 180^\circ = 462.95^\circ$$

This is equivalent to $462.95^\circ - 360^\circ = 102.95^\circ$. A shift of 180° along the orbit moves the perihelion to the opposite side of the orbit. This new position is, by definition, the location of the old aphelion, the point on the orbit farthest from the Sun (section 3.1). The current date of aphelion is July 4 (Figure 3.2). Therefore, by shifting the perihelion by 180° , the new date of perihelion becomes the old date of aphelion; so, the new date of perihelion is July 4 (Note: Figure 3.10 states that it shifts to July 3 – this small one-day difference is an artifact of the Gregorian calendar with its leap years trying to map onto an anomalistic year that is 365.26 days long).

Part 3.2

Plot insolation for each day of year and averaged over the year.

```
1 # Calculate insolation with omega shifted by 180 degrees
2 _, _, insolation_new, _ = \
3     calculate_insolation_for_year(omega_deg=282.95 + 180.0)
4
5 # Daily Insolation with New Perihelion
6 plt.subplots(figsize=(10, 6))
7 plt.title('Daily Insolation (New Perihelion at \u03D6 = 102.95\degree$)')
8 levels = np.arange(0, 551, 25)
```

```

9 cf = plt.contourf(day_of_year, lats, insolation_new, levels=levels, cmap='jet',
    extend='max')
10 plt.colorbar(cf, label='Insolation ( $W m^{-2}$ )')
11 setup_month_ticks(plt.gca(), days_in_year)

```

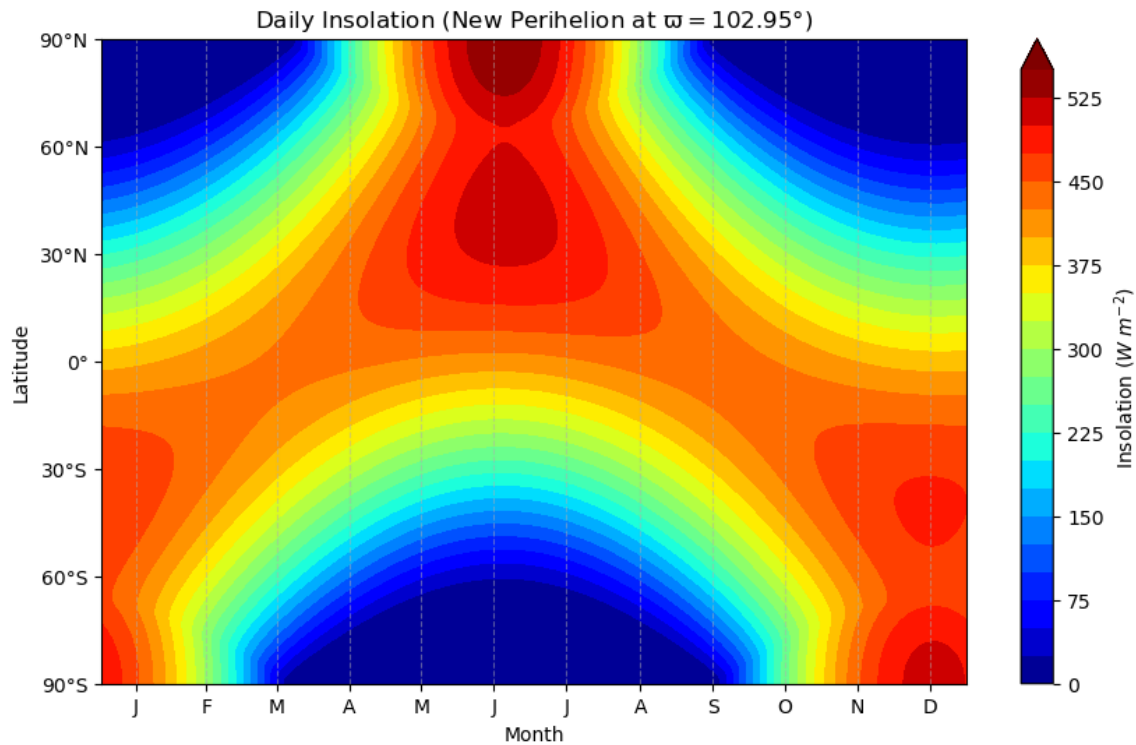


Figure 3: Daily insolation calculated with the longitude of perihelion shifted by 180° , moving perihelion from January to July (old aphelion).

```

1 # Annual Average (Original vs. New)
2 annual_avg_original = np.mean(insolation, axis=1)
3 annual_avg_new = np.mean(insolation_new, axis=1)
4
5 plt.subplots(figsize=(4, 7))
6 plt.plot(annual_avg_original, lats,
7         label='Original ( $\omega = 282.95^\circ$ )', linestyle='--')
8 plt.plot(annual_avg_new, lats, label='New ( $\omega = 102.95^\circ$ )',
9         linestyle='--')
10 plt.xlabel('Annually Averaged Insolation ( $W m^{-2}$ )')
11 plt.ylabel('Latitude')
12 plt.yticks([-90, -60, -30, 0, 30, 60, 90],
13           ['90\u00B0S', '60\u00B0S', '30\u00B0S', '0\u00B0', '30\u00B0N',
14           '60\u00B0N', '90\u00B0N'])
15 plt.ylim(-90, 90)
16 plt.grid(True, linestyle='--', alpha=0.7)
17 plt.legend()
18 plt.title('Annually Averaged Insolation Comparison')
19 plt.tight_layout()

```

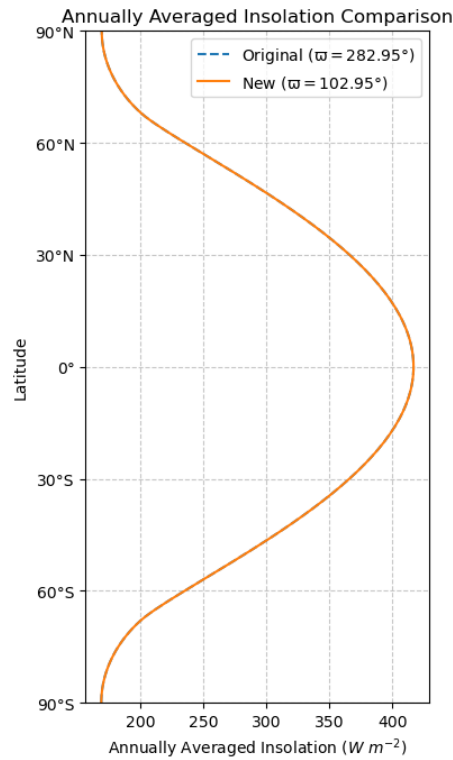


Figure 4: Comparison of annually averaged insolation, showing that a 180° shift in perihelion has no effect on the annual mean.

Part 3.3

Make a plot of the differences between the insolation for each day of the year.

```

1 # Daily Insolation Difference (New - Original)
2 plt.subplots(figsize=(10, 6))
3 plt.title('Insolation Difference: New ( $\omega = 102.95^\circ$ ) - \
4 Original ( $\omega = 282.95^\circ$ )')
5 # Use a diverging colormap (Blue-White-Red) and center it on zero
6 diff_levels = np.linspace(-80, 80, 21)
7 cf3 = plt.contourf(day_of_year, lats, insolation_new - insolation,
8                    levels=diff_levels, cmap='bwr', extend='both')
9 plt.colorbar(cf3, label='Insolation Difference ( $W m^{-2}$ )')
10 setup_month_ticks(plt.gca(), days_in_year)

```

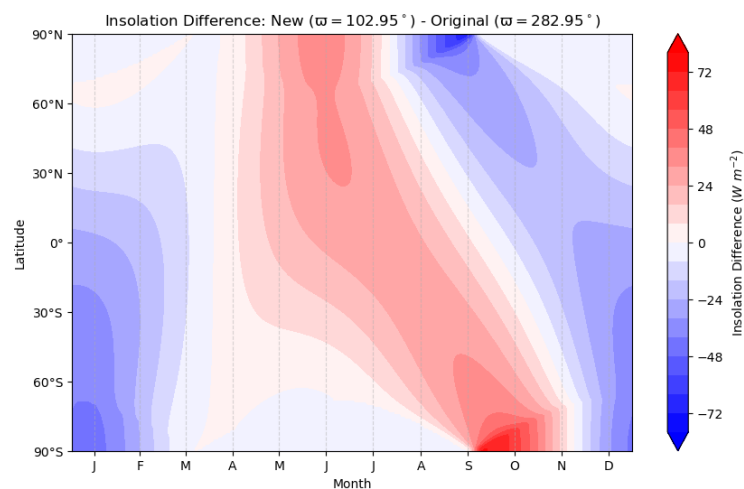


Figure 5: Daily insolation difference (180° shift - present), illustrating increased seasonal contrast in the Northern Hemisphere and decreased contrast in the Southern Hemisphere.

```

1 # Annually Averaged Insolation Difference (New - Original)
2 plt.figure(figsize=(3, 5))
3 plt.plot(np.mean(insolation_new - insolation, axis=1), lats)
4 plt.suptitle('Annually Averaged Insolation Difference')
5 plt.title('New ( $\varpi = 102.95^\circ$ ) - Original ( $\varpi = 282.95^\circ$ )')
6 plt.xlabel('Annually Averaged Insolation Difference ( $W m^{-2}$ )')
7 plt.ylabel('Latitude')
8 plt.yticks([-90, -60, -30, 0, 30, 60, 90],
9            ['90\u00B0S', '60\u00B0S', '30\u00B0S', '0\u00B0', '30\u00B0N',
10            '60\u00B0N', '90\u00B0N'])
11 plt.ylim(-90, 90)
12 plt.xlim(-1, 10)
13 plt.grid(True, linestyle='--', alpha=0.7)

```

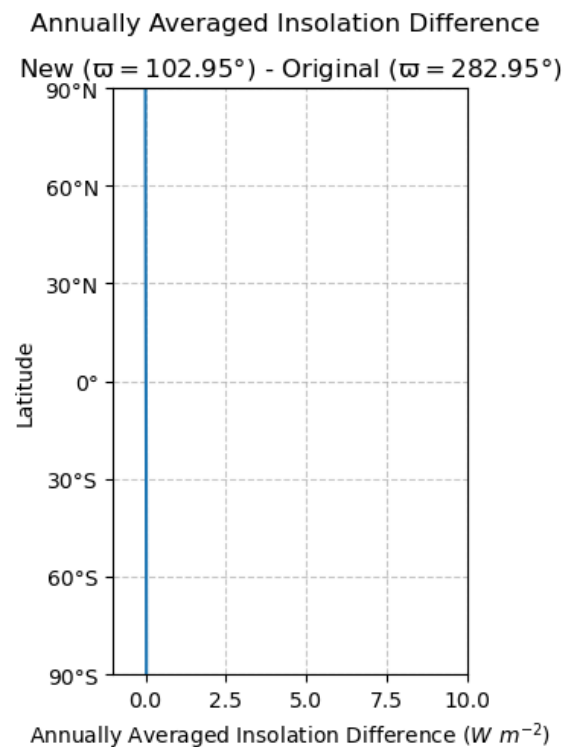


Figure 6: Annually averaged insolation difference (180° perihelion shift - present), demonstrating no net change at any latitude.

Part 3.4

What are the principal differences to the situations in parts a and b? How do they arise?

Principal Differences in Daily Insolation

Figure 3 shows the principal differences:

- **Northern Hemisphere:** Receives **more** insolation during its summer (May-August) and **less** insolation during its winter (November-February).
- **Southern Hemisphere:** Receives **less** insolation during its summer (November-February) and **more** insolation during its winter (May-August).

How they arise: This is the effect of **precession** (*Physics of Earth's Climate*, section 3.5.3).

1. In our original setup (**part 1**), perihelion (closest to sun, max radiation) occurred during Northern Hemisphere (NH) winter. This made NH winters milder and summers (at aphelion) cooler.
2. In our new setup (**part 3**), perihelion now occurs during NH summer. This makes NH summers hotter (more intense insolation) and NH winters (now at aphelion, farthest from sun) colder.

This increases the seasonal contrast in the Northern Hemisphere and, conversely, decreases the seasonal contrast in the Southern Hemisphere.

Principal Differences in Annually Averaged Insolation

Figure 4 shows the annual averages for both the original and new scenarios. We see that there is no difference as we see in **Figure 5**. The two lines in **Figure 4** are identical.

How they arise (or why they don't): A change in precession (the longitude of perihelion) has no effect on the annually averaged insolation at any latitude (**section 3.5.3**). This is due to a “cosmic coincidence” explained in **section 3.3.2**:

1. When the Earth is closer to the sun (near perihelion), it receives more intense radiation (per the inverse square law, $S \propto d^{-2}$).
2. However, at the same time, it is moving faster along its orbit (per Kepler's second law, $\dot{A} \propto d^{-2}$).

These two effects, both proportional to d^{-2} , exactly cancel each other out when averaging the insolation over a full year (**section 3.3.2**). Therefore, the total energy received by each latitude over a year remains the same, regardless of when perihelion occurs.

Part 4.1

Now reset the longitude of perihelion to its present value and decrease the obliquity to 20° . Make a plot of the differences.

```
1 # Calculate new low-obliquity insolation
2 _, _, insolation_new_obliquity, _ = \
3     calculate_insolation_for_year(gamma_deg=20.0)
4
5 difference_daily_obliquity = insolation_new_obliquity - insolation
6 plt.subplots(figsize=(10, 6))
7 plt.title('Insolation Difference (Obliquity 20.0 - 23.44)')
8 diff_levels = np.linspace(-80, 80, 21)
9 cf = plt.contourf(day_of_year, lats, difference_daily_obliquity,
10                  levels=diff_levels, cmap='bwr', extend='both')
11 plt.colorbar(cf, label='Insolation Difference ( $W m^{-2}$ )')
12 setup_month_ticks(plt.gca(), days_in_year)
```

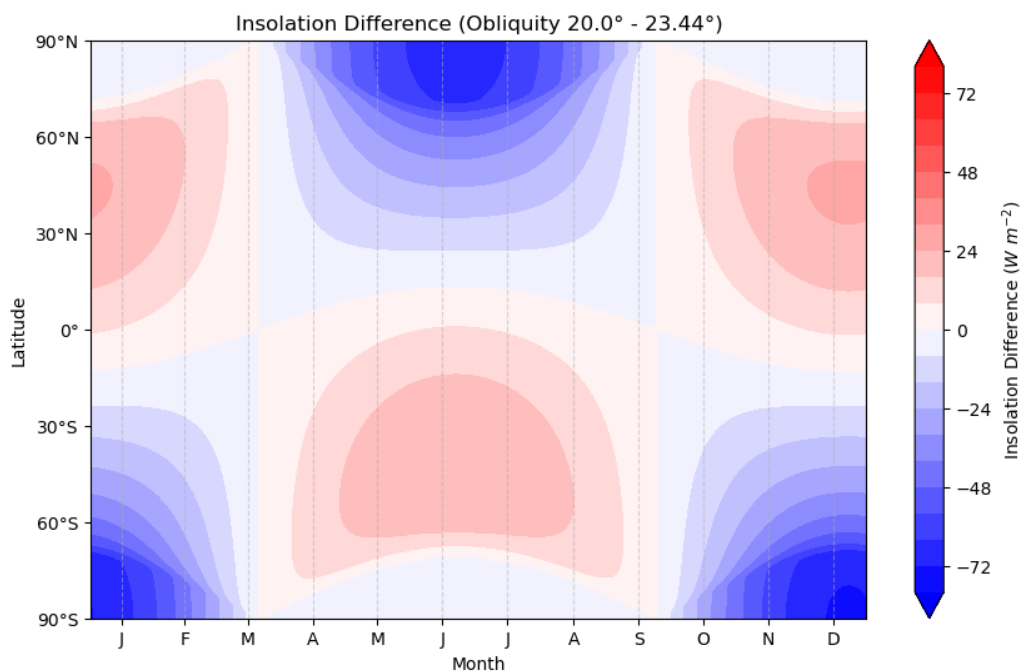


Figure 7: Daily insolation difference (20° obliquity - present), showing reduced seasonality (cooler summers, warmer winters) at high latitudes.

```

1 # Annual Average (Original vs. New)
2 annual_avg_original = np.mean(insolation, axis=1)
3 annual_avg_new_obliquity = np.mean(insolation_new_obliquity, axis=1)
4
5 fig2, ax2 = plt.subplots(figsize=(5, 7))
6 ax2.plot(annual_avg_original, lats, label='Original ( $\gamma = 23.44^\circ$ )',
7         linestyle='--')
8 ax2.plot(annual_avg_new_obliquity, lats, label='New ( $\gamma = 20.0^\circ$ )',
9         linestyle='--')
10 ax2.set_xlabel('Annually Averaged Insolation ( $W m^{-2}$ )')
11 ax2.set_ylabel('Latitude')
12 ax2.set_yticks([-90, -60, -30, 0, 30, 60, 90],
13               ['90\u00B0S', '60\u00B0S', '30\u00B0S', '0\u00B0', '30\u00B0N',
14               '60\u00B0N', '90\u00B0N'])
15 ax2.set_ylim(-90, 90)
16 ax2.grid(True, linestyle='--', alpha=0.7)
17 ax2.legend()
18 ax2.set_title('Annually Averaged Insolation Comparison')
19 fig2.tight_layout()

```

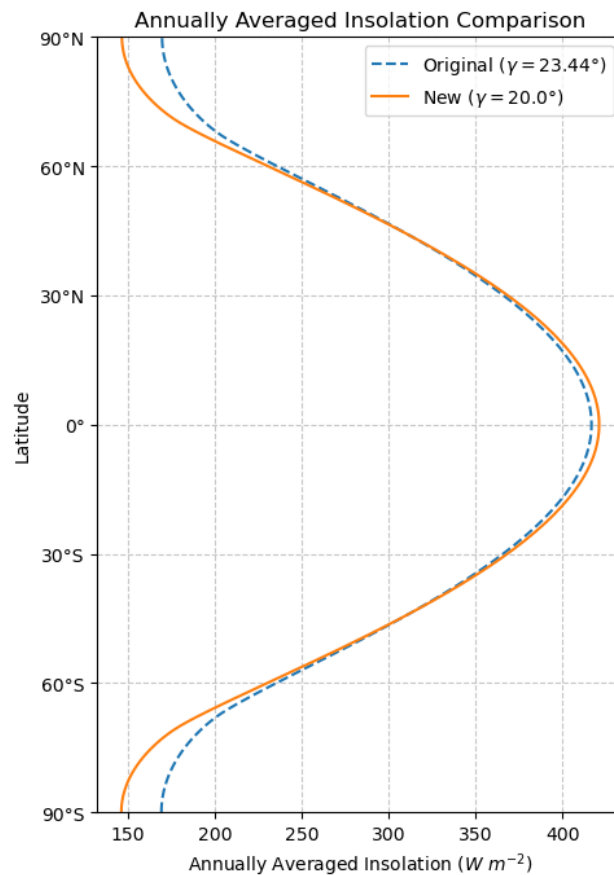


Figure 8: Comparison of annually averaged insolation (20° obliquity vs. present), showing a steeper equator-to-pole gradient with lower obliquity.

```

1 # Annual Average Insolation Difference (New - Original)
2 plt.figure(figsize=(3, 7))
3 plt.plot(np.mean(insolation_new_obliquity - insolation, axis=1), lats)
4 plt.xlabel('Annually Averaged Insolation Difference ( $W m^{-2}$ )')
5 plt.ylabel('Latitude')
6 plt.title('New ( $\gamma = 20^\circ$ ) - Old ( $\gamma = 23.44^\circ$ )')
7 plt.suptitle('Annually Averaged Insolation Difference')
8 plt.yticks([-90, -60, -30, 0, 30, 60, 90],
9           ['90\u00B0S', '60\u00B0S', '30\u00B0S', '0\u00B0', '30\u00B0N',
10           '60\u00B0N', '90\u00B0N'])
11 plt.ylim(-90, 90)
12 plt.xlim(-30, 10)
13 plt.grid(True, linestyle='--', alpha=0.7)

```

Annually Averaged Insolation Difference

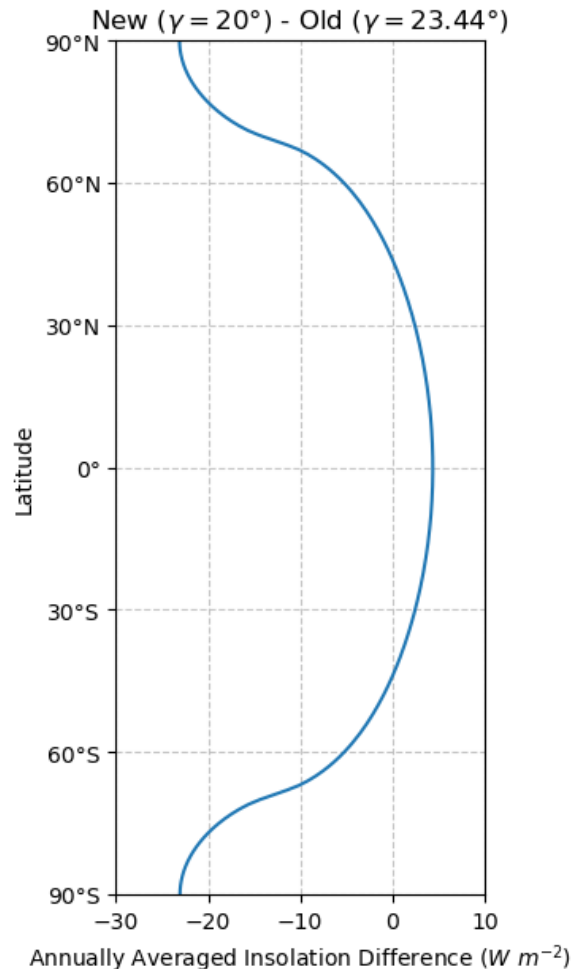


Figure 9: Annually averaged insolation difference (20° obliquity - present), highlighting increased insolation at the equator and decreased insolation at the poles.

Part 4.2

What are the principal differences to the situation in parts a and b now, and how do they arise?

Figures 7, 8, 9 from part 4.1 show the changes resulting from decreasing Earth's obliquity (γ) from 23.44° to 20.0° .

Principal Differences in Daily Insolation

Figure 7 shows a clear seasonal and latitudinal pattern:

- **High Latitudes (Poles):** Receive **less** insolation during their respective summers (blue areas) and **more** insolation during their winters (red areas).
- **Low Latitudes (Tropics):** See the opposite, but weaker, effect: **more** insolation year-round, especially during their winter season.

Overall, the summer-winter insolation contrast is **decreased** in both hemispheres.

How they arise: Obliquity is the primary driver of the seasons.

1. A lower obliquity (20.0°) means the Earth's axis is "straighter" (less tilted) relative to its orbit.
2. This reduces the maximum poleward excursion of the subsolar point. The Tropics of Cancer and Capricorn would move from $\pm 23.44^\circ$ to $\pm 20.0^\circ$.

3. Because the pole is less tilted toward the sun in summer, high-latitude summer insolation decreases. Because it is less tilted away from the sun in winter, winter insolation increases.
4. This change represents a **reduction in seasonality** (milder summers, milder winters) for the high latitudes. This is the inverse of the effect shown in *Physics of Earth's Climate*, **Figure 3.10a**, which shows an increase in obliquity.

Principal Differences in Annually Averaged Insolation

The annual average plot shows the change in the mean latitudinal distribution of energy:

- **Poles:** Receive **less** annual-mean insolation.
- **Equator:** Receives **more** annual-mean insolation.

This change makes the equator-to-pole insolation gradient **steeper**.

How they arise: Obliquity controls the annual-mean distribution of insolation between the latitudes.

1. A lower obliquity concentrates more solar energy at the equator and less at the poles over the course of a year.
2. Conversely, a high obliquity “flattens” this gradient, sending more annual-mean energy to the poles.
3. Therefore, decreasing the obliquity from 23.44° to 20.0° enhances the equator-to-pole insolation contrast, as seen in the plot.

Part 5

Earth's longitude of perihelion and obliquity vary almost periodically, with periods of about 21 kyr and 40 kyr. Given your results in the previous two exercises, how do you think can precession of the longitude of perihelion and/or obliquity variations lead to the inception and termination of glacial periods? The following paper is a fascinating discussion of ice ages as non-linear oscillators paced by Milankovich cycles and has some possible answers: <https://doi.org/10.1029/2005PA001241>.

Based on the results from **parts 3 and 4**, these orbital variations can strongly influence the seasonal insolation at high latitudes, which is the key mechanism of the Milankovitch hypothesis for ice ages (*Physics of Earth's Climate*, **section 3.6**). The central idea is that glacial periods are triggered by

weak summer insolation in the Northern Hemisphere, which allows winter snow to survive the summer melt season and accumulate into ice sheets (**section 3.6**). Here is how our results show this:

1. **Obliquity Variations (~41 kyr cycle):** Our results from **part 4** showed that **low obliquity** (a less tilted axis) leads to **less** insolation at the poles during the summer. This creates cooler summers, which would favor glacial inception. Conversely, **high obliquity** (as seen in **Fig. 3.10a** in the textbook) creates hotter summers, which would favor deglaciation. This 41 kyr cycle in obliquity is a strong match for the 41 kyr glacial cycles seen in the climate record before 1 million years ago (**section 3.6**).
2. **Precession (~21 kyr cycle):** Our results from **part 3** showed that precession controls the **timing** of the seasons relative to perihelion. Glacial inception (cool NH summers) is favored when the Northern Hemisphere summer occurs at **aphelion** (farthest from the sun). This is our “present” configuration. Conversely, glacial termination (deglaciation) is favored when NH summer occurs at **perihelion** (closest to the sun, as in our **part 3** experiment), causing intensely hot summers that rapidly melt ice sheets.
3. **Nonlinearity (The 100-kyr Problem):** Our results also showed that precession has **no effect** on the annual-mean insolation, and the effect of eccentricity (which modulates the precession effect) is very small. This suggests the climate system is not responding linearly. The provided paper explains that the 100-kyr cycles are likely an internal, nonlinear response of the climate system that gets “paced” or “phase-locked” to the orbital cycles (**Tziperman et al.**). For example, a deglaciation might only be triggered when a strong precession-driven summer warming also lines up with a high-obliquity period, causing the system to “skip” several cycles and creating a longer, 100-kyr-like period (**section 3.6**).

Part 6.1

Repeat part 4 (same longitude of perhelion as present, different obliquity) with an obliquity of 60° . Make a plot of the differences.

```
1 # Calculating new high-obliquity insolation
2 _, _, insolation_new_obliquity_60, _ = \
3     calculate_insolation_for_year(gamma_deg=60.0)
4
5 # Daily Insolation Difference (New - Original)
6 plt.subplots(figsize=(10, 6))
7 plt.title('Insolation Difference (Obliquity 60.0 - 23.44)')
8 # The changes are much more extreme now
9 diff_levels = np.linspace(-600, 600, 21)
10 cf = plt.contourf(day_of_year, lats, insolation_new_obliquity_60 - insolation,
11                  levels=diff_levels, cmap='bwr', extend='both')
12 plt.colorbar(cf, label='Insolation Difference ( $W m^{-2}$ )')
13 setup_month_ticks(plt.gca(), days_in_year)
```

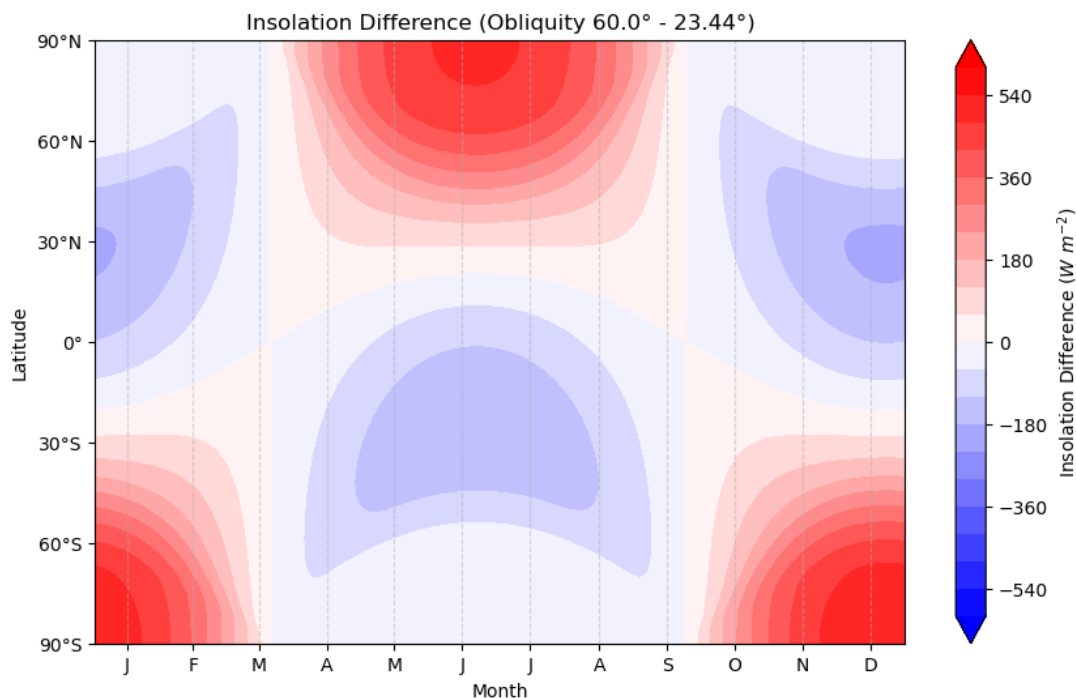


Figure 10: Daily insolation difference (60° obliquity - present), showing a dramatic increase in high-latitude summer insolation and extreme seasonality.

```
1 # Annual Average (Original vs. New)
2 annual_avg_original = np.mean(insolation, axis=1)
3 annual_avg_new_obliquity_60 = np.mean(insolation_new_obliquity_60, axis=1)
4
5 fig2, ax2 = plt.subplots(figsize=(5, 7))
6 ax2.plot(annual_avg_original, lats, label='Original ( $\gamma = 23.44^\circ$ )',
7         linestyle='--')
8 ax2.plot(annual_avg_new_obliquity_60, lats,
9         label='New ( $\gamma = 60.0^\circ$ )', linestyle='--')
10 ax2.set_xlabel('Annually Averaged Insolation ( $W m^{-2}$ )')
11 ax2.set_ylabel('Latitude')
12 ax2.set_yticks([-90, -60, -30, 0, 30, 60, 90],
13               ['90°S', '60°S', '30°S', '0°', '30°N', '60°N', '90°N'])
14
15 ax2.set_ylim(-90, 90)
16 ax2.grid(True, linestyle='--', alpha=0.7)
17 ax2.legend()
18 ax2.set_title('Annually Averaged Insolation Comparison')
19 fig2.tight_layout()
```

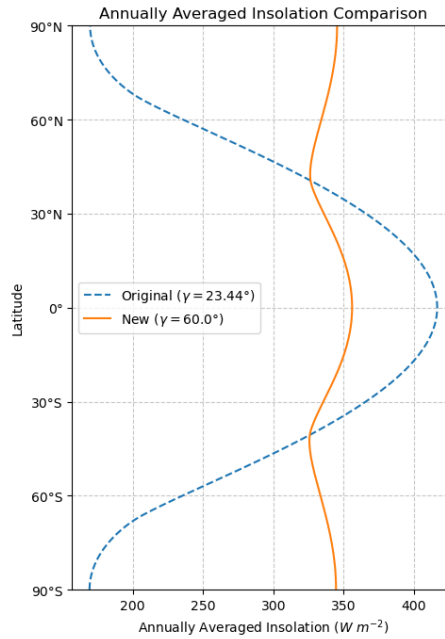


Figure 11: Comparison of annually averaged insolation (60° obliquity vs. present), revealing a new insolation maximum at the poles.

```

1 # Annual Average Insolation Difference (New - Original)
2 plt.figure(figsize=(5, 7))
3 plt.plot(np.mean(insolation_new_obliquity_60 - insolation, axis=1), lats)
4 plt.xlabel('Ann. Avg. Insolation Diff. ( $W m^{-2}$ )')
5 plt.ylabel('Latitude')
6 plt.title('New ( $\gamma = 60^\circ$ ) - Old ( $\gamma = 23.44^\circ$ )')
7 plt.yticks([-90, -60, -30, 0, 30, 60, 90],
8            ['90\u00B0S', '60\u00B0S', '30\u00B0S', '0\u00B0', '30\u00B0N',
9            '60\u00B0N', '90\u00B0N'])
10 plt.ylim(-90, 90)
11 plt.xlim(-150, 250) # Adjusted limits for the new range
12 plt.grid(True, linestyle='--', alpha=0.7)
13 plt.tight_layout()

```

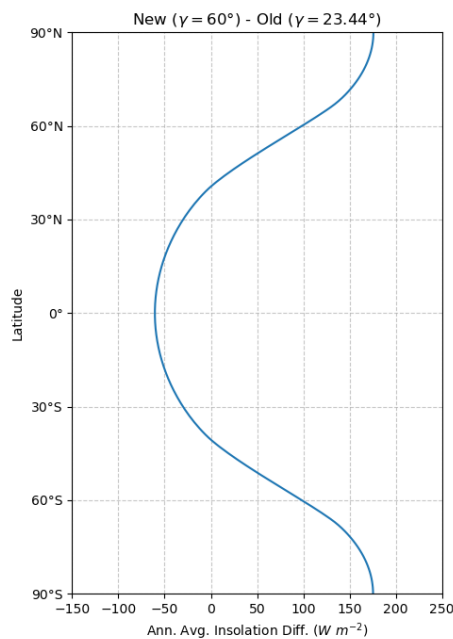


Figure 12: Annually averaged insolation difference (60° obliquity - present), demonstrating the reversal of the equator-to-pole gradient.

What are the principal differences to the situation in parts a and b now, and how do they arise? What new phenomenon arises and why?

Principal Differences in Daily Insolation

Figure 10 shows a massive amplification of the pattern from **part 4**:

- **High Latitudes (Poles):** Receive **dramatically more** insolation during their respective summers (up to $+500 \text{ W m}^{-2}$) and **less** insolation during their winters.
- **Low Latitudes (Tropics):** Receive **less** insolation year-round, especially during the summer and winter solstices.

Overall, the summer-winter insolation contrast, or **seasonality**, is **extremely increased** for all latitudes outside the new tropics.

How they arise: A high obliquity (60.0°) means the Earth's axis is tilted very far over.

1. The Tropics of Cancer and Capricorn move from $\pm 23.44^\circ$ to $\pm 60.0^\circ$.
2. The Arctic and Antarctic Circles move from $\pm 66.6^\circ$ to $\pm 30.0^\circ$.
3. This means that all latitudes poleward of 30° (i.e. Pasadena, CA) would now experience 24-hour polar day in summer and 24-hour polar night in winter, leading to extreme seasonal swings.

The New Phenomenon: Annual Average Reversal

The most significant new phenomenon appears in **Figure 11** and **12**.

- **Poles:** Receive **more** annual-mean insolation than the equator.
- **Equator:** Receives **less** annual-mean insolation.

The new phenomenon is a complete reversal of the annual-mean, equator-to-pole insolation gradient. The poles become the region of maximum annual insolation, and the equator becomes a minimum.

How it arises:

1. With a 60° tilt, the poles are exposed to 24-hour sunlight for half the year. Even though they have 24-hour darkness for the other half, the total energy received during the summer is so immense that it results in a higher annual average than any other latitude.
2. The equator, conversely, never receives direct 90° overhead sun at the solstices (the subsolar point is at 60°N or 60°S). Its average solar elevation angle over the year is now lower than that of the poles.
3. This aligns with the textbook, which states that for obliquity greater than the critical value of $\approx 54^\circ$, annual-mean insolation at the poles exceeds that at the equator (**section 3.5.2**).

Part 7

Uranus has an obliquity of 97.86° . Where is the annually averaged insolation maximal on Uranus?

The annually averaged insolation on Uranus is maximal **at its poles**. This is a more extreme case of the phenomenon discovered in **part 6**.

1. In **part 6**, we found that increasing the obliquity to 60° (which is past the critical value of $\approx 54^\circ$ mentioned in the textbook) caused the annually averaged insolation to be highest at the poles, reversing the Earth's present-day equator-to-pole gradient.
2. Uranus's obliquity of 97.86° is even further past this critical value. With such a high tilt, Uranus "effectively orbits the Sun lying on its side." This means its summer pole points almost directly at the Sun, receiving intense, continuous 24-hour insolation for a large portion of its orbit.

3. As a result, the total annual energy received by the poles is far greater than that received at the equator, which experiences two “summers” (when the Sun crosses the equator) but also two “winters” (at the solstices, when the poles are facing the Sun).
4. The textbook confirms this directly: “Annual-mean insolation on Uranus is greatest at the poles and has a minimum at the equator” (**section 3.5.2**).